

H524 Introduction to Biostatistics

Week 8: Contingency Tables and Chi-Square Tests

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Outline

- 1 Review: Where We've Been
- 2 Chi-Square Goodness-of-Fit Test (Ch 15.1)
- 3 Contingency Tables (Ch 15.2-15.3)
- 4 Fisher's Exact Test (Ch 15.4)
- 5 Effect Sizes and Measures of Association (Ch 16)
- 6 Practical Considerations and R Implementation
- 7 AI as a Learning Tool for Categorical Data
- 8 Summary and Next Steps

Quick Review: Weeks 6-7 Recap

Recent Topics Covered:

- **Week 6:** Hypothesis Testing Framework
 - One-sample and two-sample t-tests
 - Paired tests
 - Type I and Type II errors, power
- **Week 7:** Nonparametric Methods
 - Sign test, Wilcoxon signed-rank test
 - Mann-Whitney U test
 - When to use nonparametric vs parametric tests

Common theme: Comparing means (continuous outcomes)

This week: Categorical outcomes and associations

From Continuous to Categorical Data

So far in H524:

- Continuous outcomes: blood pressure, cholesterol, pain scores
- Methods: t-tests, confidence intervals, ANOVA
- Questions: Are means different between groups?

New challenge: Categorical outcomes

- Binary: disease/no disease, treatment success/failure
- Nominal: blood type (A, B, AB, O), race/ethnicity
- Ordinal: disease stage (I, II, III, IV), pain level (none, mild, moderate, severe)
- Count data: number of deaths, adverse events, cases

Key Questions:

- Is there an association between two categorical variables?
- Do observed frequencies match expected frequencies?
- Are two categorical variables independent?

Introduction to Chi-Square Tests

The chi-square (χ^2) test family:

- Tests for categorical data
- Based on comparing observed vs expected frequencies
- Named after the chi-square distribution

Two main types:

- 1 **Goodness-of-fit test:** Does a single categorical variable follow a specified distribution?
 - Example: Are blood types distributed as expected (45% O, 40% A, 11% B, 4% AB)?
- 2 **Test of independence:** Are two categorical variables associated?
 - Example: Is smoking status associated with lung cancer?

Reading: Pagano & Gauvreau Chapters 15-16

Goodness-of-Fit Test: The Basic Idea

Question: Do observed data match a theoretical distribution?

Setup:

- One categorical variable with k categories
- Null hypothesis specifies expected proportions: $p_1, p_2, \dots, p_k$
- Observed counts: $O_1, O_2, \dots, O_k$ (total n)
- Expected counts: $E_i = np_i$ for each category

Chi-square test statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Intuition:

- If observed $\approx$ expected, then χ^2 is small
- If observed differs from expected, then χ^2 is large
- Large $\chi^2 \Rightarrow$ reject H_0 (poor fit)

Chi-Square Distribution

Properties of the χ^2 distribution:

- Right-skewed, non-negative values only
- Determined by degrees of freedom: $df = k - 1$
- Mean = df , Variance = $2 \times df$
- As df increases, approaches normal distribution

Assumptions for chi-square tests:

- 1 Random sample
- 2 Independent observations
- 3 Expected counts ≥ 5 in each category
 - If violated: combine categories or use Fisher's exact test
- 4 Sufficiently large sample size

R function: `pchisq(q, df, lower.tail=FALSE)` for p-values

Example 1: Blood Type Distribution

Scenario: U.S. population blood type distribution:

- Type O: 45%
- Type A: 40%
- Type B: 11%
- Type AB: 4%

Question: Does a sample of 200 hospital patients match this distribution?

Observed counts:

Blood Type	O	A	B	AB
Observed	85	78	28	9

Hypotheses:

- H_0 : Hospital patients have same blood type distribution as general population
- H_A : Distribution differs from general population

Example 1: Calculations

Step 1: Calculate expected counts

Expected: $E_i = n \times p_i$ where $n = 200$

Blood Type	O	A	B	AB
Observed (O_i)	85	78	28	9
Expected (E_i)	90	80	22	8

Step 2: Calculate chi-square statistic

$$\begin{aligned}\chi^2 &= \frac{(85 - 90)^2}{90} + \frac{(78 - 80)^2}{80} + \frac{(28 - 22)^2}{22} + \frac{(9 - 8)^2}{8} \\ &= \frac{25}{90} + \frac{4}{80} + \frac{36}{22} + \frac{1}{8} \\ &= 0.278 + 0.050 + 1.636 + 0.125 = 2.089\end{aligned}$$

Example 1: Conclusion

Step 3: Find p-value

- Degrees of freedom: $df = k - 1 = 4 - 1 = 3$
- Test statistic: $\chi^2 = 2.089$
- P-value: $P(\chi^2_3 > 2.089) = 0.554$

R code:

```
pchisq(2.089, df=3, lower.tail=FALSE)  
# Returns: 0.554
```

Conclusion:

- P-value = 0.554 > 0.05
- Fail to reject H_0
- No evidence that hospital blood type distribution differs from general population

Example 2: Mendelian Genetics

Scenario: Mendel's pea plant experiments predict 3:1 ratio of dominant to recessive traits.

Observed data: 315 dominant, 108 recessive (total = 423)

Expected proportions: 75% dominant, 25% recessive

Calculations:

Trait	Dominant	Recessive
Observed	315	108
Expected	317.25	105.75

$$\chi^2 = \frac{(315 - 317.25)^2}{317.25} + \frac{(108 - 105.75)^2}{105.75} = 0.016 + 0.048 = 0.064$$

Result: $df = 1$, $p\text{-value} = 0.800$

Conclusion: Data consistent with 3:1 ratio

Contingency Tables: Introduction

Contingency table (cross-tabulation): Displays frequency distribution of two categorical variables

General $r \times c$ table:

- r rows (levels of variable 1)
- c columns (levels of variable 2)
- Cell counts: O_{ij} = observed count in row i , column j
- Row totals, column totals, grand total n

Key Question: Are the two variables independent?

Hypotheses:

- H_0 : Row and column variables are independent (no association)
- H_A : Row and column variables are associated (dependent)

2×2 Contingency Table

Most common case: Two binary variables

	Disease	No Disease	Total
Exposed	a	b	$a + b$
Not Exposed	c	d	$c + d$
Total	$a + c$	$b + d$	n

Examples:

- Exposure (smoking) vs Disease (lung cancer)
- Treatment (drug vs placebo) vs Outcome (success vs failure)
- Risk factor (hypertension) vs Event (stroke vs no stroke)

Measures we can calculate:

- Relative risk
- Odds ratio
- Chi-square test for independence

Chi-Square Test of Independence

Test statistic for $r \times c$ table:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Expected count formula:

$$E_{ij} = \frac{(\text{Row } i \text{ total}) \times (\text{Column } j \text{ total})}{\text{Grand total}}$$

Degrees of freedom:

$$df = (r - 1) \times (c - 1)$$

Interpretation:

- Expected counts assume independence (null hypothesis)
- Large differences between O_{ij} and $E_{ij} \Rightarrow$ evidence of association
- Compare χ^2 to chi-square distribution with appropriate df

Example 3: Smoking and Lung Cancer

Case-control study: 100 lung cancer patients, 100 controls

	Lung Cancer	No Cancer	Total
Smoker	85	55	140
Non-smoker	15	45	60
Total	100	100	200

Question: Is there an association between smoking and lung cancer?

Hypotheses:

- H_0 : Smoking status and lung cancer are independent
- H_A : Smoking status and lung cancer are associated

Example 3: Expected Counts

Calculate expected counts under independence:

$$E_{ij} = \frac{(\text{Row total}) \times (\text{Column total})}{n}$$

	Cancer	No Cancer
Smoker	$\frac{140 \times 100}{200} = 70$	$\frac{140 \times 100}{200} = 70$
Non-smoker	$\frac{60 \times 100}{200} = 30$	$\frac{60 \times 100}{200} = 30$

Summary table:

Cell	Observed	Expected
Smoker + Cancer	85	70
Smoker + No Cancer	55	70
Non-smoker + Cancer	15	30
Non-smoker + No Cancer	45	30

Example 3: Chi-Square Calculation

Calculate test statistic:

$$\begin{aligned}\chi^2 &= \frac{(85 - 70)^2}{70} + \frac{(55 - 70)^2}{70} + \frac{(15 - 30)^2}{30} + \frac{(45 - 30)^2}{30} \\ &= \frac{225}{70} + \frac{225}{70} + \frac{225}{30} + \frac{225}{30} \\ &= 3.214 + 3.214 + 7.500 + 7.500 = 21.43\end{aligned}$$

Find p-value:

- $df = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$
- $\chi^2 = 21.43$
- $P\text{-value} = P(\chi_1^2 > 21.43) < 0.0001$

Conclusion: Very strong evidence of association between smoking and lung cancer ($p < 0.0001$)

Example 4: Treatment Efficacy

Clinical trial: New drug vs placebo for treating infection

	Cured	Not Cured	Total
Drug	48	12	60
Placebo	32	28	60
Total	80	40	120

Expected counts:

	Cured	Not Cured
Drug	40	20
Placebo	40	20

$$\chi^2 = \frac{(48 - 40)^2}{40} + \frac{(12 - 20)^2}{20} + \frac{(32 - 40)^2}{40} + \frac{(28 - 20)^2}{20} = 9.60$$

Result: $df = 1$, $p\text{-value} = 0.002$

Conclusion: Strong evidence drug is more effective than placebo ▶

Larger Contingency Tables

Example: 3×3 table - Disease severity by age group

	Mild	Moderate	Severe	Total
Young (< 40)	45	20	5	70
Middle (40-60)	30	35	15	80
Older (> 60)	15	25	10	50
Total	90	80	30	200

Analysis:

- $df = (3 - 1)(3 - 1) = 4$
- Calculate 9 expected counts
- Sum contributions: $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
- Compare to χ^2_4 distribution

Interpretation: Significant result indicates association between age and disease severity, but doesn't specify pattern

When Chi-Square Fails: Small Samples

Problem: Chi-square test assumes large sample sizes

- Rule of thumb: All expected counts ≥ 5
- When violated, chi-square approximation poor
- P-values unreliable

Solution: Fisher's Exact Test

- Calculates exact probability under null hypothesis
- No large-sample approximation needed
- Works for any sample size
- Most commonly used for 2×2 tables

Trade-offs:

- More computationally intensive
- Exact p-values (conservative)
- R implementation: `fisher.test()`

Fisher's Exact Test: Example

Scenario: Rare disease study with small sample

	Disease	No Disease	Total
Exposed	8	2	10
Not Exposed	3	7	10
Total	11	9	20

Check chi-square assumptions:

- Expected count (Exposed, No Disease) = $\frac{10 \times 9}{20} = 4.5$ (OK)
- Expected count (Not Exposed, Disease) = $\frac{10 \times 11}{20} = 5.5$ (OK)
- Borderline for chi-square assumptions

Better approach: Use Fisher's exact test

R code: `fisher.test(matrix(c(8,2,3,7), nrow=2))`

Result: P-value = 0.035 (two-sided)

Fisher's Exact Test: How It Works

Concept: Calculate probability of observed table and all more extreme tables

For 2×2 table:

	Col 1	Col 2	Total
Row 1	a	b	$a + b$
Row 2	c	d	$c + d$
Total	$a + c$	$b + d$	n

Exact probability (hypergeometric):

$$P = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}}$$

P-value: Sum probabilities for observed and all more extreme configurations (fixing margins)

Beyond Significance: Measuring Association

Chi-square test tells us:

- Whether association exists (p-value)
- But NOT the strength of association

Effect size measures for 2×2 tables:

- ① **Relative Risk (RR):** Ratio of probabilities

$$RR = \frac{P(\text{Disease} \mid \text{Exposed})}{P(\text{Disease} \mid \text{Not Exposed})} = \frac{a/(a+b)}{c/(c+d)}$$

- ② **Odds Ratio (OR):** Ratio of odds

$$OR = \frac{a \times d}{b \times c}$$

- ③ **Risk Difference:** Absolute difference in probabilities

$$RD = \frac{a}{a+b} - \frac{c}{c+d}$$

Relative Risk vs Odds Ratio

When to use which?

Relative Risk:

- Cohort studies, randomized trials
- When incidence can be calculated
- Direct interpretation: "X times more likely"
- Example: $RR = 2.5$ means exposed group is 2.5 times more likely to develop disease

Odds Ratio:

- Case-control studies
- When incidence cannot be calculated directly
- Approximates RR when disease is rare
- Symmetric: $OR(A | B) = OR(B | A)$
- Used in logistic regression

Interpretation:

Example: Calculating RR and OR

Smoking and lung cancer data (from earlier):

	Cancer	No Cancer	Total
Smoker	85	55	140
Non-smoker	15	45	60
Total	100	100	200

Relative Risk:

$$RR = \frac{85/140}{15/60} = \frac{0.607}{0.250} = 2.43$$

Odds Ratio:

$$OR = \frac{85 \times 45}{55 \times 15} = \frac{3825}{825} = 4.64$$

Interpretation:

- Smokers are 2.43 times more likely to have lung cancer
- Odds of cancer among smokers are 4.64 times odds among

Confidence Intervals for RR and OR

Point estimates alone insufficient - need uncertainty!

95% CI for Odds Ratio:

$$CI = \exp [\log(OR) \pm 1.96 \times SE_{\log(OR)}]$$

where

$$SE_{\log(OR)} = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

Smoking example: $OR = 4.64$

$$SE_{\log(OR)} = \sqrt{\frac{1}{85} + \frac{1}{55} + \frac{1}{15} + \frac{1}{45}} = 0.330$$

$$95\% CI = \exp[\log(4.64) \pm 1.96 \times 0.330] = (2.40, 8.98)$$

Interpretation: We're 95% confident true OR is between 2.40 and 8.98

Key: CI doesn't include 1 $\Rightarrow$ significant association

McNemar's Test for Paired Data

Special case: Matched pairs or repeated measures with binary outcomes

Example: Before/after treatment study

	After: Pos	After: Neg	Total
Before: Pos	a	b	$a + b$
Before: Neg	c	d	$c + d$
Total	$a + c$	$b + d$	n

McNemar's test statistic:

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

Only discordant pairs (b and c) contribute!

When to use:

- Matched case-control studies
- Before-after studies with same subjects

Chi-Square Tests in R

Goodness-of-fit test:

```
# Observed counts
observed <- c(85, 78, 28, 9)
# Expected proportions
expected_props <- c(0.45, 0.40, 0.11, 0.04)

# Chi-square test
chisq.test(observed, p=expected_props)
```

Test of independence:

```
# Create contingency table
data <- matrix(c(85, 55, 15, 45), nrow=2, byrow=TRUE)
rownames(data) <- c("Smoker", "Non-smoker")
colnames(data) <- c("Cancer", "No Cancer")
```

```
# Chi-square test
```

Fisher's Exact Test and Effect Sizes in R

Fisher's exact test:

```
# For 2x2 table
data <- matrix(c(8, 2, 3, 7), nrow=2)
fisher.test(data)
```

Odds ratio and confidence interval:

```
# Using fisher.test
result <- fisher.test(data)
result$estimate      # Odds ratio
result$conf.int      # 95% CI

# Manual calculation
OR <- (data[1,1] * data[2,2]) / (data[1,2] * data[2,1])
```

Relative risk (manual):

```
RR <- (data[1,1] / (data[1,1] + data[1,2])) / (data[2,1] / (data[2,1] + data[2,2]))
```

Visualizing Contingency Tables

```
# Create data
```

```
data <- matrix(c(85, 55, 15, 45), nrow=2)
rownames(data) <- c("Smoker", "Non-smoker")
colnames(data) <- c("Cancer", "No Cancer")
```

```
# Mosaic plot
```

```
mosaicplot(data, main="Smoking and Lung Cancer",
            color=c("red", "blue"),
            xlab="Smoking Status", ylab="Cancer Status")
```

```
# Grouped barplot
```

```
barplot(data, beside=TRUE, legend=TRUE,
        col=c("darkred", "darkgreen"),
        main="Smoking and Lung Cancer",
        xlab="Cancer Status", ylab="Count")
```

```
# Proportions plot
```

Common Mistakes and Pitfalls

1. Using chi-square with small expected counts

- Check all expected counts ≥ 5
- If violated: use Fisher's exact test or combine categories

2. Confusing association with causation

- Chi-square shows association, not causation
- May be confounding variables

3. Using chi-square for paired/matched data

- Independence assumption violated
- Use McNemar's test instead

4. Ignoring effect size

- Significant p-value doesn't mean strong association
- Always report OR or RR with confidence intervals

5. Multiple testing without adjustment

- Testing many pairs in large tables inflates Type I error

Choosing the Right Test

Situation	Test
One categorical variable vs expected distribution	Chi-square goodness-of-fit
Two independent categorical variables (large sample)	Chi-square test of independence
Two independent categorical variables (small sample)	Fisher's exact test
Paired/matched binary data	McNemar's test
Ordinal variables with natural ordering	Consider Cochran-Armitage trend test
Multiple categorical variables	Log-linear models (beyond this course)

AI for Contingency Table Analysis

How AI can help this week:

- **Test selection:** Choosing between chi-square, Fisher's, McNemar's
- **Expected counts:** Calculating and verifying assumptions
- **Effect sizes:** Computing RR, OR, and confidence intervals
- **Interpretation:** Understanding what results mean
- **Visualization:** Creating mosaic plots and grouped bar charts
- **R code:** Implementing tests and extracting results

Example prompts:

- "Show me how to do chi-square test in R for a 3x3 contingency table"
- "Calculate odds ratio and 95% CI from a 2x2 table with counts: 25, 10, 5, 20"
- "When should I use Fisher's exact test instead of chi-square?"
- "Create a mosaic plot for my contingency table data"

AI Demo: Chi-Square Test Analysis

Prompt: "Analyze this 2x2 table and compute chi-square test, RR, and OR"

Claude AI Response:

```
# Enter data
data <- matrix(c(48, 12, 32, 28), nrow=2, byrow=TRUE)
rownames(data) <- c("Drug", "Placebo")
colnames(data) <- c("Cured", "Not Cured")

# Display table
print(data)
addmargins(data)

# Chi-square test
chi_result <- chisq.test(data)
cat("Chi-square statistic:", round(chi_result$statistic, 3), "\n")
cat("P-value:", format.pval(chi_result$p.value, digits=3), "\n")

# Expected counts
cat("\nExpected counts:\n")
print(round(chi_result$expected, 2))

# Effect sizes
# Relative Risk
rr <- (data[1,1]/sum(data[1,])) / (data[2,1]/sum(data[2,]))
cat("\nRelative Risk:", round(rr, 3), "\n")

# Odds Ratio
or <- (data[1,1]*data[2,2]) / (data[1,2]*data[2,1])
cat("Odds Ratio:", round(or, 3), "\n")
```

AI Demo: Visualizing Contingency Tables

Prompt: "Create informative visualizations for my contingency table"

Claude AI Response:

```
# Data
data <- matrix(c(85, 55, 15, 45), nrow=2)
rownames(data) <- c("Smoker", "Non-smoker")
colnames(data) <- c("Cancer", "No Cancer")

# Create 2x2 plot layout
par(mfrow=c(2,2))

# 1. Mosaic plot
mosaicplot(data, main="Mosaic Plot", color=c("red", "lightblue"),
            xlab="Smoking", ylab="Cancer")

# 2. Grouped barplot (counts)
barplot(data, beside=TRUE, legend=TRUE, col=c("darkred", "darkgreen"),
        main="Counts by Group", ylab="Frequency",
        args.legend=list(x="topright"))

# 3. Stacked proportions
prop_col <- prop.table(data, margin=2)
barplot(prop_col, col=c("darkred", "darkgreen"), legend=TRUE,
        main="Proportions (by Column)", ylab="Proportion",
        args.legend=list(x="topright"))

# 4. Side-by-side proportions (risk)
prop_row <- prop.table(data, margin=1)
barplot(prop_row, beside=TRUE, legend=TRUE, col=c("darkred", "darkgreen"),
        main="Cancer Rate by Smoking Status", ylab="Proportion",
        names.arg=c("Cancer", "No Cancer"),
```

AI Demo: Interpreting Results

Prompt: "Explain the results of my chi-square test and what the OR means"

Claude AI Response:

Your chi-square test results:

- Chi-square statistic: 21.43
- Degrees of freedom: 1
- P-value: < 0.0001

Interpretation:

1. Statistical Significance:

The extremely small p-value (< 0.0001) provides very strong evidence against the null hypothesis of independence.

We reject the null and conclude there IS a significant association between smoking and lung cancer.

2. Odds Ratio = 4.64 (95% CI: 2.40, 8.98):

- Smokers have 4.64 times the ODDS of having lung cancer compared to non-smokers

AI Best Practices for Categorical Analysis

DO:

- Verify assumptions (expected counts)
- Check which test is appropriate
- Request both p-values and effect sizes
- Ask for visualization code
- Request interpretation in context
- Double-check calculations

DON'T:

- Trust test choice blindly
- Skip assumption checking
- Report only p-values
- Confuse RR and OR
- Ignore small expected counts
- Conclude causation from association

Key Verification Steps:

- 1 Check table margins sum correctly
- 2 Verify expected counts ≥ 5 for chi-square
- 3 Confirm OR and RR calculations manually
- 4 Ensure CI interpretation is correct
- 5 Validate that conclusions match the data

AI Task: Power Analysis for Contingency Tables

Advanced prompt: "Calculate sample size needed to detect $OR=2.5$ "

This week's AI-integrated tasks:

- 1 Generate R code for chi-square test with multiple tables
- 2 Calculate and interpret odds ratios with confidence intervals
- 3 Create publication-quality mosaic plots
- 4 Explain when to use Fisher's vs chi-square
- 5 Simulate contingency table data with specified association

Sample Problem:

"I have a 3x4 contingency table of treatment group (3 levels) by outcome (4 levels). Create R code to:

- 1) Perform chi-square test
- 2) Visualize the table
- 3) Calculate standardized residuals to identify which cells contribute most to significance
- 4) Interpret results in clinical context"

Summary: Categorical Data Analysis

Key Concepts:

- **Chi-square goodness-of-fit:** One variable vs expected distribution
- **Chi-square independence test:** Association between two categorical variables
- **Fisher's exact test:** Small samples or when chi-square assumptions violated
- **McNemar's test:** Paired/matched binary data
- **Effect sizes:** Relative risk (RR) and odds ratio (OR)

R Functions:

- `chisq.test()` – Chi-square tests
- `fisher.test()` – Fisher's exact test
- `mcnemar.test()` – McNemar's test
- Manual calculations for RR, OR, and CIs

Remember:

- Check assumptions (especially expected counts)

Next Week (Week 9): Correlation and Simple Linear Regression

- Quantifying linear relationships between continuous variables
- Pearson and Spearman correlation
- Simple linear regression
- Prediction and inference

Connection to this week:

- This week: Association between categorical variables
- Next week: Association between continuous variables
- Both involve testing independence/association
- Different methods, same statistical thinking

Reading: Pagano & Gauvreau Chapter 17 (Correlation and Regression)

Thank you!

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This week's materials:

- Lab: Chi-square and contingency table exercises in R
- Worksheet: Practice problems with partners
- Office hours: Available for questions on categorical analysis